Combination

Derive an equation of displacement vs time for a free-falling object from height h at initial velocity of u.

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$$v = u - gt \quad \text{and} \quad s = \left(\frac{u+v}{2}\right)t$$

Comments: From definitions.

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$$s = \left(\frac{u+(u-gt)}{2}\right)t$$

Comments: Distribute the terms.

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$$v = u - gt \quad \text{and} \quad s = \left(\frac{u+v}{2}\right)t$$

$$s = \left(\frac{u+(u-gt)}{2}\right)t \quad \Rightarrow \quad s = ut - \frac{1}{2}gt^2$$

Comments: That's it.